

Finite-Dimensional Constraint Filters and Their Path-Integral Extension

Coarea Localization, Normalization, and the MTT Source Boundary

Peter Nero

July 2026, Version 1

Abstract

Delta functionals occur throughout path-integral physics: as hard constraints, gauge-fixing conditions, phase-space reductions, Lagrange-multiplier insertions, and exact selection of histories. In standard formalism they are usually treated as distributional devices. This paper continues the delta/projection program by showing that a delta functional may be understood as the zero-width limit of a finite admissibility filter over histories. The rigorous core is finite-dimensional: for a smooth constraint map $C : M \rightarrow \mathbb{R}^m$ with 0 a regular value, normalized Gaussian filters

$$\mathcal{K}_\varepsilon(C(x)) = (2\pi\varepsilon^2)^{-m/2} \exp(-|C(x)|^2/2\varepsilon^2)$$

converge, as measures, to the surface delta $\delta(C(x)) dx$, including the standard Jacobian factor from the coarea formula. The infinite-dimensional path-integral statement is then treated as a formal extension requiring regularization. In conditional MTT language, the hard delta $\delta[C]$ is the singular limit of a normalized finite admissibility weight $\mathcal{N}_\varepsilon \exp[-J_{\text{adm}}/\varepsilon^2]$. This also clarifies the relation between multiplier representations, large-penalty localization, and finite admissibility tubes, and unifies the interpretation of classical constraint shells, gauge-fixing deltas, and path-integral restrictions: exact constraint enforcement is the zero-width idealization of finite admissibility selection.

Version 1 Revision Note

Supersedes The unversioned April 2026 manuscript on path-integral constraint filters.

Reason The rigorous coarea theorem was sound, but several later formulas suppressed the normalization required for a delta limit and the carrier assignment could be read as already derived.

Resolution This version proves that unnormalized penalties converge to zero, restores the normalization and induced Jacobian, and separates the finite-dimensional theorem from the regulator-dependent functional-integral and MTT source problems.

Retained result Normalized approximate identities converge to the coarea surface delta, and multiplier damping is the Fourier-dual finite-width description.

Remaining boundary A regulated functional measure, anomaly and quotient control, and a selected MTT admissibility cost are still required in a physical field theory.

Contents

1 Purpose and claim discipline

2

2	From hard constraints to finite filters	3
3	Finite-dimensional theorem	3
3.1	Gaussian constraint filters	4
4	General admissibility kernels	4
5	Path integrals: formal functional extension	5
5.1	Scope warning	5
6	Relation to Lagrange multipliers	6
7	Large-penalty localization	6
8	Admissible filters versus arbitrary penalties	7
9	Relation to gauge fixing	7
10	MTT interpretation	7
10.1	Triadic reading	8
11	Soft constraints versus hard constraints	8
12	Diagnostics and finite-width effects	9
13	Scope of proof	9
14	Conclusion	10

1 Purpose and claim discipline

The previous papers in this sequence reinterpreted several recurring delta functions: identity kernels, point sources, gauge slices, measurement projectors, contact vertices, conservation deltas, spectral peaks, white noise, and microcanonical shells. The present paper treats the path-integral version:

$$\delta[C[\phi]].$$

The guiding claim is:

$\delta[C] = \text{zero-width limit of a finite admissibility filter over histories.}$

This paper has two layers.

Layer	Status
Finite-dimensional constraint theorem	rigorous, proved using the coarea formula
Functional-integral extension	formal; requires regulator, measure choice, and anomaly control
MTT interpretation	structural; finite admissibility weights replace hard delta constraints

The paper does not claim to define all path integrals nonperturbatively. It isolates one mathematical pattern: hard functional deltas are singular limits of finite-width constraint filters.

2 From hard constraints to finite filters

A standard constrained path integral has the schematic form

$$Z = \int \mathcal{D}\phi \, \delta[C[\phi]] \, e^{iS[\phi]/\hbar}.$$

The delta functional enforces exact membership in the constraint surface $C[\phi] = 0$.

The finite-filter replacement is

$$\delta[C[\phi]] \rightsquigarrow \mathcal{K}_\varepsilon[C[\phi]] = \mathcal{N}_\varepsilon \exp\left(-\frac{1}{2\varepsilon^2} \|C[\phi]\|^2\right),$$

where the normalization depends on the regulated constraint space and the norm is chosen by an admissibility metric. In m finite constraint dimensions with the Euclidean metric, $\mathcal{N}_\varepsilon = (2\pi\varepsilon^2)^{-m/2}$. The finite-filter path integral is then

$$Z_\varepsilon = \int \mathcal{D}\phi \, \mathcal{K}_\varepsilon[C[\phi]] \, e^{iS[\phi]/\hbar}.$$

In Euclidean signature one may instead write

$$Z_\varepsilon^E = \int \mathcal{D}\phi \, \exp\left(-S_E[\phi] - \frac{1}{2\varepsilon^2} \|C[\phi]\|^2\right).$$

This is a soft-constraint or penalty formulation. The hard constraint is recovered only in the $\varepsilon \downarrow 0$ limit after the appropriate normalization is included.

3 Finite-dimensional theorem

Let M be an n -dimensional smooth Riemannian manifold with volume measure $d\mu$. Let

$$C : M \rightarrow \mathbb{R}^m, \quad m \leq n,$$

be a smooth constraint map, and assume 0 is a regular value. Then

$$\Sigma := C^{-1}(0)$$

is a smooth submanifold of codimension m .

Define the constraint Jacobian

$$J_C(x) := \sqrt{\det(DC_x DC_x^*)},$$

where $DC_x : T_x M \rightarrow \mathbb{R}^m$ and the adjoint is taken with respect to the Riemannian metric on M and the Euclidean metric on $\mathbb{R}^m$.

The surface delta is defined by

$$\int_M f(x) \delta(C(x)) d\mu(x) = \int_\Sigma \frac{f(x)}{J_C(x)} d\sigma(x),$$

for test functions f , where $d\sigma$ is the induced hypersurface measure.

3.1 Gaussian constraint filters

Let

$$k_\varepsilon(u) := (2\pi\varepsilon^2)^{-m/2} \exp(-|u|^2/2\varepsilon^2).$$

Theorem 3.1 (Gaussian constraint filters converge to surface deltas). *Let $f \in C_c^\infty(M)$. Under the regular-value assumption above,*

$$\lim_{\varepsilon \downarrow 0} \int_M f(x) k_\varepsilon(C(x)) d\mu(x) = \int_\Sigma \frac{f(x)}{J_C(x)} d\sigma(x).$$

Equivalently,

$$k_\varepsilon(C(x)) d\mu(x) \longrightarrow \delta(C(x)) d\mu(x)$$

weakly as measures.

Proof. By the coarea formula, for any integrable h ,

$$\int_M h(x) J_C(x) d\mu(x) = \int_{\mathbb{R}^m} \left(\int_{C^{-1}(u)} h(x) d\sigma_u(x) \right) du.$$

Apply this with

$$h(x) = \frac{f(x) k_\varepsilon(C(x))}{J_C(x)},$$

which is well-defined on a neighborhood of Σ because 0 is a regular value and f has compact support. Then

$$\int_M f(x) k_\varepsilon(C(x)) d\mu(x) = \int_{\mathbb{R}^m} k_\varepsilon(u) \left[\int_{C^{-1}(u)} \frac{f(x)}{J_C(x)} d\sigma_u(x) \right] du.$$

For u near 0, the bracketed expression is smooth by the regular-level-set theorem and compact support of f . Denote it by $F(u)$. Since k_ε is an approximate identity on $\mathbb{R}^m$,

$$\int_{\mathbb{R}^m} k_\varepsilon(u) F(u) du \rightarrow F(0).$$

But

$$F(0) = \int_\Sigma \frac{f(x)}{J_C(x)} d\sigma(x).$$

This proves the claim. □

4 General admissibility kernels

Gaussian filters are convenient but not essential. Let

$$k_\varepsilon(u) = \varepsilon^{-m} k(u/\varepsilon),$$

where $k \in C_c^\infty(\mathbb{R}^m)$, $k \geq 0$, and $\int k(u) du = 1$. The same proof gives:

$$k_\varepsilon(C(x)) d\mu(x) \rightarrow \delta(C(x)) d\mu(x).$$

Thus the hard delta does not depend on the detailed shape of the finite filter. The detailed shape controls finite- ε corrections. In MTT language, those corrections encode finite admissibility width.

Proposition 4.1 (Normalization is indispensable). *For $m > 0$, the unnormalized weights*

$$u \longmapsto e^{-|u|^2/(2\varepsilon^2)}$$

converge to zero as distributions on $\mathbb{R}^m$, not to δ_0 . Multiplication by $(2\pi\varepsilon^2)^{-m/2}$, or an equivalent normalization for the chosen kernel, is required for a unit-mass delta limit.

Proof. For every compactly supported test function f ,

$$\left| \int_{\mathbb{R}^m} f(u) e^{-|u|^2/(2\varepsilon^2)} du \right| \leq \|f\|_{\infty} (2\pi)^{m/2} \varepsilon^m \longrightarrow 0.$$

The normalized family has unit integral and is an approximate identity. $\square$

5 Path integrals: formal functional extension

The formal field-theoretic analogue is obtained by replacing $x \in M$ with a field configuration $\phi \in \mathcal{F}$ and $C(x)$ with a functional constraint

$$C[\phi] \in \mathcal{Y}.$$

A hard constrained functional integral has the schematic form

$$Z = \int_{\mathcal{F}} \mathcal{D}\phi \, \delta(C[\phi]) e^{iS[\phi]/\hbar}.$$

The finite-filter version is

$$Z_{\varepsilon} = \int_{\mathcal{F}} \mathcal{D}\phi \, \mathcal{K}_{\varepsilon}(C[\phi]) e^{iS[\phi]/\hbar},$$

where

$$\mathcal{K}_{\varepsilon}(C) = \mathcal{N}_{\varepsilon} \exp\left(-\frac{1}{2\varepsilon^2} \|C\|_{\mathcal{Y}}^2\right).$$

In Euclidean form:

$$Z_{\varepsilon}^E = \int_{\mathcal{F}} \mathcal{D}\phi \, \exp\left(-S_E[\phi] - \frac{1}{2\varepsilon^2} \|C[\phi]\|_{\mathcal{Y}}^2\right).$$

5.1 Scope warning

In infinite dimensions, the symbols

$$\mathcal{D}\phi, \quad \delta(C[\phi]), \quad \det(DC \, DC^*)$$

are formal unless a regulator, Gaussian measure, lattice approximation, pAQFT construction, or other definition is supplied. The rigorous theorem of this paper is finite-dimensional. The functional-integral formulas are structural extensions of the same mechanism.

6 Relation to Lagrange multipliers

A delta constraint may also be represented by a Fourier integral:

$$\delta(C) = \int \frac{d\lambda}{2\pi} e^{i\lambda C}.$$

For multiple constraints:

$$\delta^{(m)}(C) = \int \frac{d^m \lambda}{(2\pi)^m} e^{i\lambda \cdot C}.$$

This expresses hard constraint enforcement by an auxiliary multiplier field. The finite-filter version corresponds to adding a damping envelope in multiplier space:

$$k_\varepsilon(C) = \int \frac{d^m \lambda}{(2\pi)^m} e^{i\lambda \cdot C} e^{-\varepsilon^2 |\lambda|^2/2}.$$

Thus finite admissibility width in constraint space is equivalent to suppressing arbitrarily large Lagrange-multiplier modes.

7 Large-penalty localization

The same limit may be expressed dynamically as a large-penalty limit. In Euclidean signature, replace the constrained integral

$$\int \delta(C(x)) e^{-S(x)} d\mu(x)$$

by

$$\int \exp \left[-S(x) - \frac{|C(x)|^2}{2\varepsilon^2} \right] d\mu(x).$$

As $\varepsilon \downarrow 0$, the penalty term suppresses all configurations outside an $O(\varepsilon)$ -tube around the constraint surface $C^{-1}(0)$. In local coordinates (u, v) , where $u = C(x) \in \mathbb{R}^m$ and v parametrizes the constraint surface, the integral has the schematic form

$$\int e^{-|u|^2/2\varepsilon^2} F(u, v) du dv.$$

After multiplication by the normalizing factor $(2\pi\varepsilon^2)^{-m/2}$, the Gaussian integral localizes to $u = 0$, giving

$$\int_{C^{-1}(0)} \frac{e^{-S(x)}}{J_C(x)} d\sigma(x),$$

with $J_C = (\det DC DC^*)^{1/2}$. Thus the finite admissibility filter is not merely a visual analogy for a delta; it has the same localizing asymptotics as the hard constrained measure.

In Lorentzian signature the corresponding statement is more delicate because the path integral is oscillatory. The penalty term may be introduced after Wick rotation, by $i0$ -type damping, or inside a regulator-defined construction. The structural point remains the same: the hard functional delta is the zero-width limit of a filter that increasingly localizes histories near $C = 0$.

8 Admissible filters versus arbitrary penalties

A finite-width constraint filter changes the theory at finite ε . Therefore the replacement

$$\delta[C] \rightsquigarrow \mathcal{K}_\varepsilon[C]$$

is not automatically legitimate as physics. It is legitimate only if the filter is either:

1. a regulator that is removed in the end;
2. an effective model for finite experimental or coarse-graining resolution; or
3. an admissibility kernel derived from the underlying MTT projection and closure-cost structure.

This distinction is important. A generic penalty may violate a symmetry, distort the measure, or select the wrong sector. In gauge theory, for example, finite gauge-tube filters must be compatible with the Faddeev–Popov/BV–BRST quotient structure if they are to define the same physical quotient in the sharp limit. In MTT terms, the filter must be an admissible projection weight, not an arbitrary smearing inserted by hand.

9 Relation to gauge fixing

Gauge fixing is a special case of constraint filtering. A gauge condition

$$G[A] = 0$$

inserts

$$\delta(G[A])$$

together with the Faddeev–Popov determinant.

The finite admissibility version is

$$\delta(G[A]) \rightsquigarrow \mathcal{K}_\varepsilon(G[A]) = \mathcal{N}_\varepsilon \exp\left(-\frac{1}{2\varepsilon^2} \|G[A]\|^2\right).$$

This replaces a hard gauge slice by a finite gauge tube. The Faddeev–Popov determinant remains the Jacobian of projection from gauge-orbit coordinates to the chosen slice, while the filter width controls how sharply the representative section is enforced.

This paper therefore generalizes the gauge paper:

gauge-fixing delta $\subset$ path-integral constraint delta $\subset$ finite admissibility filter limit.

10 MTT interpretation

In Modal Triplet Theory, admissible descriptions are not arbitrary configurations. They are configurations that survive projection, stabilization, and coherence constraints. Therefore a hard functional delta

$$\delta(C[\phi])$$

should not be treated as primitive. It is the zero-width limit of an admissibility filter:

$$\mathcal{K}_\varepsilon[C[\phi]] = \mathcal{N}_\varepsilon \exp\left(-\frac{1}{\varepsilon^2} J_{\text{adm}}[\phi]\right),$$

where J_{adm} measures the degree of constraint violation, closure strain, gauge-slice distance, or failure of coherent continuation. The normalization and the induced Jacobian must be defined by the regulated measure; they are not optional decorations.

The hard limit is

$$\mathcal{K}_\epsilon[C[\phi]] \longrightarrow \delta(C[\phi])$$

only in a regulator-defined weak sense. The finite-dimensional theorem proves the model for this statement; it does not by itself construct the functional measure.

10.1 Triadic reading

The same delta may involve different MTT carrier roles.

Carrier role	Constraint-filter interpretation
Circle C	exact bookkeeping closure, return consistency, conservation shells
Lens L	representative selection inside redundancy classes, gauge slices
Nil N	survivor-basin thresholds, admissibility boundaries, collapse-like selection

This is a proposed carrier dictionary, not a derivation of the three assignments from the shared-circle geometry. A path-integral delta functional may be read as downstream notation for an admissibility condition only after the relevant source map and quotient measure have been supplied.

11 Soft constraints versus hard constraints

The finite filter

$$\exp(-\|C\|^2/2\epsilon^2)$$

does not merely approximate a mathematical delta. For finite ϵ , it defines a different effective theory: one in which constraint violation is penalized rather than forbidden.

This distinction matters.

Regime	Mathematical form	MTT reading
Hard constraint	$\delta(C)$	zero-width admissibility limit
Soft constraint	$e^{-\ C\ ^2/2\epsilon^2}$	finite admissibility tolerance
Penalty action	$S + \ C\ ^2/2\epsilon^2$	closure-cost deformation
Unconstrained	no filter	no enforced admissibility shell

The hard limit may be useful and often correct. But MTT treats it as a limit, not as the primitive description.

12 Diagnostics and finite-width effects

Replacing $\delta[C]$ by $\mathcal{K}_\varepsilon[C]$ suggests concrete finite-width diagnostics:

1. constraint surfaces become finite admissibility tubes;
2. gauge slices become finite gauge tubes;
3. hard classical shells become soft shells;
4. path histories with small closure violation contribute with controlled suppression;
5. infinitely sharp Lagrange-multiplier modes are damped;
6. exact constraint enforcement is recovered only as $\varepsilon \downarrow 0$.

The physical relevance of these corrections depends on the sector. In ordinary effective field theory they may be only regulator artifacts. In MTT they may encode finite coherence capacity.

13 Scope of proof

Statement	Status
Gaussian filters on regular finite-dimensional constraint surfaces converge to surface deltas	proved
General approximate-identity filters give the same surface delta	proved by same coarea argument
Large-penalty filters localize to the constraint surface in the sharp limit	proved locally by the same coarea/Gaussian argument
Functional delta filters behave analogously in path integrals	formal, regulator-dependent
Gauge-fixing deltas are a special case of constraint filtering	structural and standard
Finite penalties preserve physics only when regulator-removable or admissibility-derived	caveat, not automatic
MTT admissibility costs generate all physical constraint filters	interpretive target, not proved here

14 Conclusion

Path-integral delta functionals are usually written as hard constraints:

$$\delta[C[\phi]].$$

This paper shows that, at the rigorous finite-dimensional level, such deltas arise as zero-width limits of finite constraint filters. The same pattern extends formally to functional integrals, where a hard delta can be replaced by a finite admissibility weight

$$\mathcal{N}_\varepsilon \exp[-\|C[\phi]\|^2/2\varepsilon^2].$$

The conditional MTT reading is:

$\delta[C] = \text{singular shadow of finite admissibility filtering.}$

This unifies classical constraint shells, gauge-fixing deltas, soft penalty actions, and path-integral restrictions under one projection-first interpretation.

References

- [1] H. Federer, *Geometric Measure Theory*, Springer (1969; reprint 1996). [doi:10.1007/978-3-642-62010-2](https://doi.org/10.1007/978-3-642-62010-2).
- [2] L. D. Faddeev and V. N. Popov, “Feynman diagrams for the Yang–Mills field,” *Physics Letters B* **25** (1967), 29–30. [doi:10.1016/0370-2693\(67\)90067-6](https://doi.org/10.1016/0370-2693(67)90067-6).
- [3] I. A. Batalin and G. A. Vilkovisky, “Gauge algebra and quantization,” *Physics Letters B* **102** (1981), 27–31. [doi:10.1016/0370-2693\(81\)90205-7](https://doi.org/10.1016/0370-2693(81)90205-7).